

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 5-(Ethoxycarbonyl)pentylzinc bromide, 0.5M in THF

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>H58304</b>                                                                                        |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Flammable liquids                                                      | Category 2 |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 |

#### Health hazards

|                                                    |              |
|----------------------------------------------------|--------------|
| Acute Oral Toxicity                                | Category 4   |
| Skin Corrosion/Irritation                          | Category 1 B |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Carcinogenicity                                    | Category 2   |
| Specific target organ toxicity - (single exposure) | Category 3   |

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark



Health Hazard



Corrosion

Signal Word

Danger

#### Hazard Statements

H225 - Highly flammable liquid and vapor  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
AUH019 - May form explosive peroxides

#### Precautionary Statements

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P223 - Do not allow contact with water  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water  
P402 + P404 - Store in a dry place. Store in a closed container  
P501 - Dispose of contents/ container to an approved waste disposal plant

#### Other information

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component                            | CAS No      | Weight % |
|--------------------------------------|-------------|----------|
| Tetrahydrofuran                      | 109-99-9    | 85.28    |
| 5-(Ethoxycarbonyl)pentylzinc bromide | 312693-01-9 | 14.72    |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.                                                                                                                |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                                                                                                                                          |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                                                                                       |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                             |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                             |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                              |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                                                                                                                                                                                                                               |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Dry sand. Carbon dioxide (CO<sub>2</sub>). Powder. Do not use water or foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen bromide, Zinc oxide.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep refrigerated. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component       | Australia                                  | New Zealand WEL                                                                                   | ACGIH TLV                            | The United Kingdom                                                                                                        | Germany                                                                                                                               |
|-----------------|--------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 100 ppm<br>TWA: 295 mg/m <sup>3</sup> | TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup><br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>Skin | TWA: 50 ppm<br>STEL: 100 ppm<br>Skin | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 150 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 |

|  |  |  |  |  |                                                                                                                                   |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  |  | Stunden). MAK<br>TWA: 150 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 300 mg/m <sup>3</sup><br>Haut |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------|

### Biological limit values

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component       | Australia | New Zealand                                                                                         | European Union | United Kingdom | Germany                                          |
|-----------------|-----------|-----------------------------------------------------------------------------------------------------|----------------|----------------|--------------------------------------------------|
| Tetrahydrofuran |           | 2 mg/g creatinine (urine)<br>end of exposure or shift,<br>within 1 hour of end of<br>exposure (THF) |                |                | Tetrahydrofuran: 2 mg/L<br>urine (end of shift ) |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material  | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|-----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber  | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Viton (R)       | recommendations   |                 |                 |                       |
| Butyl rubber    |                   |                 |                 |                       |
| Neoprene gloves |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

#### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

#### Recommended half mask:-

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Yellow - Brown - Black   |                                             |
| <b>Physical State</b>                          | Liquid                   |                                             |
| <b>Odor</b>                                    | No information available |                                             |
| <b>Odor Threshold</b>                          | No data available        |                                             |
| <b>pH</b>                                      | No information available |                                             |
| <b>Melting Point/Range</b>                     | No data available        |                                             |
| <b>Softening Point</b>                         | No data available        |                                             |
| <b>Boiling Point/Range</b>                     | 66 °C / 150.8 °F         |                                             |
| <b>Flash Point</b>                             | -17 °C / 1.4 °F          | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available        |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available        |                                             |
| <b>Vapor Pressure</b>                          | 23 hPa @ 20 °C           |                                             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.98 g/cm3               | @ 20 °C                                     |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | Immiscible               |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Tetrahydrofuran                                | 0.45                     |                                             |
| <b>Autoignition Temperature</b>                | No data available        |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | No data available        |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available |                                             |
| <b>Other information</b>                       |                          |                                             |
| <b>Molecular Formula</b>                       | C8 H15 BrO2 Zn           |                                             |
| <b>Molecular Weight</b>                        | 288.49                   |                                             |

## Section 10 - Stability and Reactivity

|                                         |                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes                                                                                    |
| <b>Stability</b>                        | Air sensitive. Water reactive. May form precipitate.                                   |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition.                      |
| <b>Incompatible Materials</b>           | Acids, Acid chlorides, Oxidizing agent.                                                |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Hydrogen bromide. Zinc oxide. |
| <b>Hazardous Polymerization</b>         | No information available.                                                              |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| Oral                       | Category 4                                                       |
| Dermal                     | Based on available data, the classification criteria are not met |
| Inhalation                 | Based on available data, the classification criteria are not met |

**Toxicology data for the components**

| Component       | LD50 Oral          | LD50 Dermal           | LC50 Inhalation                               |
|-----------------|--------------------|-----------------------|-----------------------------------------------|
| Tetrahydrofuran | 1650 mg/kg ( Rat ) | > 2000 mg/kg (Rabbit) | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                             | Test method                                       | Test species | Study result    |
|---------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( 85.28 ) | Local Lymph Node Assay OECD<br>Test Guideline 429 | mouse        | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                             | Test method                                             | Test species          | Study result |
|---------------------------------------|---------------------------------------------------------|-----------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( 85.28 ) | OECD Test Guideline 476<br>Gene cell mutation           | in vivo<br>Mammalian  | negative     |
|                                       | OECD Test Guideline 473<br>Chromosomal aberration assay | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; Category 2

Limited evidence of a carcinogenic effect The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component       | Australia | New Zealand             | New South<br>Wales | Western<br>Australia | IARC     | EU | UK | Germany |
|-----------------|-----------|-------------------------|--------------------|----------------------|----------|----|----|---------|
| Tetrahydrofuran |           | Suspected<br>carcinogen |                    |                      | Group 2B |    |    |         |

(g) reproductive toxicity; No data available

| Component                             | Test method             | Test species / Duration | Study result      |
|---------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( 85.28 ) | OECD Test Guideline 416 | Rat 2 Generation        | NOAEL = 3,000 ppm |

(h) STOT-single exposure; Category 3

Results / Target organs

Respiratory system  
Central nervous system (CNS)

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Section 12 - Ecological Information**



|                                              |                                                                                        |                                                                                                                                                                                              |                                                |                 |
|----------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------|
| <b>Ecotoxicity effects</b>                   |                                                                                        | May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.                                                                            |                                                |                 |
| <b>Component</b>                             | <b>Freshwater Fish</b>                                                                 | <b>Water Flea</b>                                                                                                                                                                            | <b>Freshwater Algae</b>                        | <b>Microtox</b> |
| Tetrahydrofuran                              | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50:<br>2820 mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h                                                                                                                                                 |                                                |                 |
| <b>Persistence and Degradability</b>         |                                                                                        | Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist.                               |                                                |                 |
| <b>Persistence</b>                           |                                                                                        | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                                                              |                                                |                 |
| <b>Degradation in sewage treatment plant</b> |                                                                                        |                                                                                                                                                                                              |                                                |                 |
| <b>Bioaccumulative Potential</b>             |                                                                                        | May have some potential to bioaccumulate                                                                                                                                                     |                                                |                 |
| <b>Component</b>                             | <b>log Pow</b>                                                                         |                                                                                                                                                                                              | <b>Bioconcentration factor (BCF)</b>           |                 |
| Tetrahydrofuran                              | 0.45                                                                                   |                                                                                                                                                                                              | No data available                              |                 |
| <b>Mobility</b>                              |                                                                                        | The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air |                                                |                 |
| <b>Endocrine Disruptor Information</b>       |                                                                                        |                                                                                                                                                                                              |                                                |                 |
| <b>Component</b>                             | <b>EU - Endocrine Disruptors Candidate List</b>                                        | <b>EU - Endocrine Disruptors - Evaluated Substances</b>                                                                                                                                      | <b>Japan - Endocrine Disruptor Information</b> |                 |
| Tetrahydrofuran                              | Group III Chemical                                                                     |                                                                                                                                                                                              |                                                |                 |
| <b>Persistent Organic Pollutant</b>          |                                                                                        | This product does not contain any known or suspected substance                                                                                                                               |                                                |                 |
| <b>Ozone Depletion Potential</b>             |                                                                                        | This product does not contain any known or suspected substance                                                                                                                               |                                                |                 |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                  |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                              |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                                             |
|--------------------------------|-------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                      |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE |
| <b>Technical Shipping Name</b> | (5-(Ethoxycarbonyl)pentylzinc bromide, TETRAHYDROFURAN)     |
| <b>Hazard Class</b>            | 4.3                                                         |
| <b>Subsidiary Hazard Class</b> | 3                                                           |
| <b>Packing Group</b>           | II                                                          |

### ADG

|                                |                                                             |
|--------------------------------|-------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                      |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE |
| <b>Technical Shipping Name</b> | (5-(Ethoxycarbonyl)pentylzinc bromide, TETRAHYDROFURAN)     |
| <b>Hazard Class</b>            | 4.3                                                         |



|                                       |                     |
|---------------------------------------|---------------------|
| <b>Subsidiary Hazard Class</b>        | 3                   |
| <b>Packing Group</b>                  | II                  |
| <b>Component</b>                      | <b>Hazchem Code</b> |
| Tetrahydrofuran<br>109-99-9 ( 85.28 ) | 2YE                 |

#### IATA

|                                |                                                             |
|--------------------------------|-------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                      |
| <b>Proper Shipping Name</b>    | Organometallic substance, liquid, water-reactive, flammable |
| <b>Technical Shipping Name</b> | (5-(Ethoxycarbonyl)pentylzinc bromide, TETRAHYDROFURAN)     |
| <b>Hazard Class</b>            | 4.3                                                         |
| <b>Subsidiary Hazard Class</b> | 3                                                           |
| <b>Packing Group</b>           | II                                                          |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## **Section 15 - Regulatory Information**

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                  | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------------|-------------------------------------------------------------|------------------------|
| Tetrahydrofuran - 109-99-9 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

| Component                  | Australian - Illicit Drug Precursors/Reagents Substance List | Chemicals of Security Concern |
|----------------------------|--------------------------------------------------------------|-------------------------------|
| Tetrahydrofuran - 109-99-9 | Category 3                                                   |                               |

#### **Legend**

Category 3 - Chemicals and apparatus that may be used in the illicit production of drugs. Purchases from this list should alert companies or organizations to seek further indicators of any suspicious orders or enquiries. No official reporting is required for items on this list unless considered warranted

National pollutant inventory Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

| Component                  | Australia | New South Wales | Western Australia | New Zealand          |
|----------------------------|-----------|-----------------|-------------------|----------------------|
| Tetrahydrofuran - 109-99-9 |           |                 |                   | Suspected carcinogen |

**International Inventories**

| Component       | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Tetrahydrofuran | X    | X     | 203-726-8 | -      | X    | X   | -    | X     | X    | X    | X     | KE-33454 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component                            | CAS No      | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------|-------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Tetrahydrofuran                      | 109-99-9    | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| 5-(Ethoxycarbonyl)pentylzinc bromide | 312693-01-9 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Revision Date** 19-Nov-2022

**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**